Host and burst properties vs axis ratio: outlier-robust statistics

Property vs axis ratio q	N	r_S (p)	AD p	bootstrap 95%	Kendall τ (p)	delete-1 range	Theil–Sen slope [95%]
DM_{host} (rest frame)	55	-0.06 (0.64)	0.37	[-0.32, +0.20]	-0.04 (0.63)	-0.12 to -0.04	-38 [-239, 159]
DM_{host}, foreground halos removed	49	-0.09 (0.56)	0.35	[-0.37, +0.20]	-0.06 (0.51)	-0.14 to -0.04	-42.6 [-275, 136]
DM_{host}, foreground excess over random sightlines removed	49	-0.09 (0.52)	0.35	[-0.36, +0.19]	-0.06 (0.57)	-0.15 to -0.05	-53.9 [-256, 144]
DM_{host}, foreground removed (all photo-z galaxies in front)	49	-0.07 (0.61)	0.35	[-0.35, +0.21]	-0.06 (0.56)	-0.13 to -0.03	-45.4 [-288, 135]
DM_{host}, spectroscopic foreground only	49	-0.06 (0.70)	0.46	[-0.34, +0.23]	-0.04 (0.70)	-0.12 to -0.02	-20.6 [-219, 172]
DM_{host}, foreground excess removed (published method)	49	-0.09 (0.53)	0.49	[-0.36, +0.18]	-0.06 (0.55)	-0.15 to -0.05	-71 [-279, 154]
log|RM|	41	-0.02 (0.91)	0.33	[-0.32, +0.29]	-0.00 (0.99)	-0.09 to +0.04	-0.00544 [-1.46, 1.12]
$\log \tau_{scatt}$	29	-0.17 (0.39)	0.65	[-0.55, +0.27]	-0.11 (0.41)	-0.28 to -0.07	-1.03 [-2.92, 1.48]
Linear polarization	24	-0.33 (0.11)	0.49	[-0.70, +0.12]	-0.24 (0.096)	-0.45 to -0.25	-0.467 [-1.08, 0.188]
A_V, old stars	30	-0.46 (0.010)	0.004	[-0.67, -0.15]	-0.28 (0.032)	-0.53 to -0.41	-0.429 [-1.06, -0.0207]
DM_{host} (rest frame), m_r < 21	45	-0.09 (0.54)	0.086	[-0.38, +0.20]	-0.07 (0.52)	-0.16 to -0.06	-54 [-273, 168]
DM_{host} (rest frame), n < 2	46	-0.04 (0.81)	0.58	[-0.33, +0.25]	-0.02 (0.85)	-0.10 to +0.00	-10.5 [-177, 168]
DM_{host} (rest frame), bulge fits removed	52	-0.07 (0.64)	0.26	[-0.34, +0.20]	-0.04 (0.64)	-0.12 to -0.04	-38.4 [-254, 168]
DM_{host}, foreground excess over random sightlines removed, m_r < 21	32	-0.11 (0.51)	0.16	[-0.43, +0.20]	-0.06 (0.57)	-0.19 to -0.06	-51 [-263, 171]
DM_{host}, foreground excess over random sightlines removed, n < 2	42	-0.07 (0.65)	0.31	[-0.38, +0.24]	-0.04 (0.72)	-0.14 to -0.02	-38.2 [-225, 159]
DM_{host}, foreground excess over random sightlines removed, bulge fits removed	46	-0.09 (0.57)	0.27	[-0.37, +0.20]	-0.05 (0.63)	-0.15 to -0.05	-44.1 [-256, 157]

Bootstrap: 10,000 resamples. Since every host has $q > q_0$, rank correlations with q and with $\cos i$ are identical (sign flipped for i).
AD p: two-sample Anderson–Darling test that y (in 'y vs x'; the property, against q) has the same distribution for hosts below and above the median of x (permutation p, about 5,000 shuffles).
A_V of the old stars is the only Spearman $p < 0.05$ case (dust in inclined disks).